

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.2	10.4	0.2	0	0	62.2
U3 - Toxicity prior Beta(0.15, 0.85)	28.7	9	0.1	0	0	62.2
U2 - Toxicity prior Beta(0.3, 0.7)	26.9	10.8	0.6	0.1	0	61.6
U3 - Toxicity prior Beta(0.3, 0.7)	29.3	8.5	0	0.1	0	62.1
U2 - Toxicity prior Beta(0.5, 0.5)	29.3	10.1	0.8	0.5	0	59.3
U3 - Toxicity prior Beta(0.5, 0.5)	31.9	7.9	0.4	0.3	0	59.5
U2 - Toxicity prior Beta(1, 1)	29.2	10.1	0.7	0.4	0	59.6
U3 - Toxicity prior Beta(1, 1)	32.3	7.7	0.3	0.2	0	59.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.4	3.7	1.3	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	16.3	3.7	1.4	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	16.2	3.8	1.4	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	16.3	3.8	1.3	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	15.9	4	1.7	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	15.9	4.1	1.7	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	15.9	4.1	1.7	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	15.8	4.1	1.7	0.5	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	1.9	0.4	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	1.9	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	1.9	0.4	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	1.9	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.5	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	1.9	0.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.7	1.9	0.5	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.7	1.9	0.5	0.2	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	36	22.8	30.5	7.5	1.3	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	39.5	23.6	26.7	7.5	0.8	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	35.6	22.7	29.9	8.6	1.3	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	38.8	23.2	26.8	8.3	1	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	34	19.5	31.5	11.5	1.6	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	35.9	21.5	30.4	9.2	1.1	1.9
U2 - Toxicity prior Beta(1, 1)	34.4	19.6	31.4	11	1.7	1.9
U3 - Toxicity prior Beta(1, 1)	36.2	21.5	29.3	9.9	1.2	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.5	6	6.9	3.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	12.5	6.1	6.7	3.2	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	12.4	5.8	6.8	3.4	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	12.4	5.7	6.9	3.4	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	11.5	5.8	7	3.9	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	11.4	5.8	7.1	3.9	1.6	
U2 - Toxicity prior Beta(1, 1)	11.6	5.8	6.9	3.9	1.5	
U3 - Toxicity prior Beta(1, 1)	11.5	5.8	7	3.9	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.6	1.2	0.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.6	1.2	0.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.5	1.2	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.5	1.2	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.4	1.4	0.8	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.4	1.4	0.8	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	2.4	1.4	0.8	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.4	1.4	0.8	0.3	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.7	46.3	17.9	6.2	0.5	10.4
U3 - Toxicity prior Beta(0.15, 0.85)	19.7	48.4	16.2	5.1	0.3	10.3
U2 - Toxicity prior Beta(0.3, 0.7)	17.9	44.2	19.7	7.4	0.5	10.3
U3 - Toxicity prior Beta(0.3, 0.7)	19	46.5	17	6.8	0.5	10.2
U2 - Toxicity prior Beta(0.5, 0.5)	17.2	42.8	20.3	8.7	1.3	9.7
U3 - Toxicity prior Beta(0.5, 0.5)	19.2	45.2	18.9	6.3	0.6	9.8
U2 - Toxicity prior Beta(1, 1)	17.1	42.4	21.1	8.7	1	9.7
U3 - Toxicity prior Beta(1, 1)	19	44.5	19.2	6.7	0.8	9.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.8	8.3	5.6	3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	10.8	8.4	5.6	3	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.6	8	5.8	3.1	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	8.1	5.7	3.1	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	8.1	5.9	3.6	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	9.8	8.1	5.9	3.6	1.4	
U2 - Toxicity prior Beta(1, 1)	9.8	8.1	6	3.6	1.4	
U3 - Toxicity prior Beta(1, 1)	9.9	8.1	5.9	3.6	1.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.1	1.1	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.1	1.1	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.1	0.9	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.1	0.8	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.3	0.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.3	0.9	0.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.3	0.9	0.3	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.4	0.9	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	9.5	23.9	44	11.9	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	13.7	11.5	20.3	42.3	10.3	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	9.7	22.1	44.2	13.3	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	12.7	10.9	20.5	42.1	11.9	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	8.7	20	47.9	15.6	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	10.7	10.1	18.6	45.9	12.8	1.9
U2 - Toxicity prior Beta(1, 1)	5.7	8.6	20.5	48.1	15.2	1.9
U3 - Toxicity prior Beta(1, 1)	10.4	9.7	18.6	46.3	13.1	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.8	4.7	7.1	8.1	4.9	
U3 - Toxicity prior Beta(0.15, 0.85)	4.8	4.8	7	8.2	4.9	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	4.6	6.7	8.5	5.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	4.6	6.9	8.4	5.3	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	4.2	6.4	9.1	5.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	4.3	6.3	9.1	5.6	
U2 - Toxicity prior Beta(1, 1)	4.4	4.2	6.4	9.2	5.7	
U3 - Toxicity prior Beta(1, 1)	4.4	4.2	6.4	9.2	5.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.5	2.2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.5	2.3	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.2	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.5	2.2	0.7	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.5	2.2	0.7	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.5	2.2	0.7	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	35.6	26.3	21.5	8	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	7.8	42.3	23.4	16.7	5.6	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	35.8	25.5	21	9	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	7.7	40.9	23.3	17.7	6.2	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	3.9	33.1	26.1	23.1	9.6	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	7.9	39.8	24.8	17.4	5.9	4.2
U2 - Toxicity prior Beta(1, 1)	4.3	33.5	26.3	21.3	10.4	4.2
U3 - Toxicity prior Beta(1, 1)	7.2	41.3	24	17.3	6	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	7	6.7	6	4.5	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	7	6.7	6	4.5	
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	6.8	6.6	6.1	4.8	
U3 - Toxicity prior Beta(0.3, 0.7)	5	6.9	6.6	6.1	4.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	6.4	6.7	6.3	5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	6.4	6.7	6.3	5.1	
U2 - Toxicity prior Beta(1, 1)	4.9	6.4	6.7	6.3	5.1	
U3 - Toxicity prior Beta(1, 1)	4.9	6.5	6.6	6.3	5.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.2	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.4	1.6	2.2	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.3	1.6	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	3.3	1.7	2.2	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.8	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.1	3.3	1.7	2.2	1	
U3 - Toxicity prior Beta(1, 1)	0.1	3.3	1.8	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	69.2	12.9	2.6	0.6	0.1	14.6
U3 - Toxicity prior Beta(0.15, 0.85)	71.7	11.4	1.8	0.3	0.1	14.7
U2 - Toxicity prior Beta(0.3, 0.7)	70	13.1	2.3	0.4	0.1	14.1
U3 - Toxicity prior Beta(0.3, 0.7)	71.1	12.5	1.4	0.4	0.1	14.5
U2 - Toxicity prior Beta(0.5, 0.5)	67.5	15	2.6	0.7	0.2	14
U3 - Toxicity prior Beta(0.5, 0.5)	67.8	14.7	2.2	0.6	0.2	14.5
U2 - Toxicity prior Beta(1, 1)	67.2	15	2.9	0.3	0.2	14.4
U3 - Toxicity prior Beta(1, 1)	69	14.2	2.4	0.6	0.1	13.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.9	4	1.7	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	20.9	4	1.7	0.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	21	4	1.8	0.6	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	20.9	4	1.7	0.6	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	20.1	4.5	1.9	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	20.1	4.4	2	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	20.1	4.5	1.9	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	20.2	4.4	1.9	0.8	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.5	0.5	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.4	1.5	0.5	0.2	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.3	44.6	17	6.8	1.9	2.4
U3 - Toxicity prior Beta(0.15, 0.85)	28.7	46.2	14.9	6.7	1.1	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	26.7	43	17.7	8.3	2.1	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.5	45.4	16.2	6.8	0.8	2.3
U2 - Toxicity prior Beta(0.5, 0.5)	26	39.5	19.2	10.9	2.1	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	27.8	42.7	17.4	8.4	1.4	2.3
U2 - Toxicity prior Beta(1, 1)	25.8	39.8	19.4	10.2	2.5	2.3
U3 - Toxicity prior Beta(1, 1)	28	42.9	16.8	8.5	1.5	2.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.8	7.8	5.2	3.2	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	11.8	7.8	5.3	3.2	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	11.7	7.6	5.3	3.2	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	11.7	7.7	5.3	3.2	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	10.8	7.7	5.7	3.5	2	
U3 - Toxicity prior Beta(0.5, 0.5)	10.8	7.8	5.5	3.6	1.9	
U2 - Toxicity prior Beta(1, 1)	10.8	7.8	5.5	3.5	2	
U3 - Toxicity prior Beta(1, 1)	10.9	7.7	5.6	3.5	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1	0.8	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1	0.9	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1	0.8	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.1	0.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.1	0.9	0.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.1	0.9	0.3	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.1	0.9	0.3	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	42.3	8.4	1.2	0.3	0	47.8
U3 - Toxicity prior Beta(0.15, 0.85)	43.3	7.8	0.7	0.2	0	48
U2 - Toxicity prior Beta(0.3, 0.7)	43.1	8.4	0.9	0.5	0	47.1
U3 - Toxicity prior Beta(0.3, 0.7)	43.9	7.8	0.6	0.4	0	47.3
U2 - Toxicity prior Beta(0.5, 0.5)	43.2	8.3	1.4	0.8	0	46.3
U3 - Toxicity prior Beta(0.5, 0.5)	44	8.1	1.2	0.5	0	46.2
U2 - Toxicity prior Beta(1, 1)	43.1	8.3	1.3	0.7	0	46.6
U3 - Toxicity prior Beta(1, 1)	43.9	7.9	0.8	0.6	0	46.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.8	3.1	1.1	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	18.8	3.1	1.1	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	18.7	3.2	1.1	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	18.7	3.2	1.2	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	18	3.7	1.4	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	18	3.8	1.4	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	18	3.7	1.4	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	18	3.7	1.4	0.4	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.5	0.3	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.5	0.3	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1.5	0.3	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1.5	0.3	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.4	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.4	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.4	0.4	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.4	1.5	0.4	0.1	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.6	17.7	16.6	11.8	15.5	19.8
U3 - Toxicity prior Beta(0.15, 0.85)	25.4	16	15.2	10.3	13.2	19.9
U2 - Toxicity prior Beta(0.3, 0.7)	17.2	16.8	17.4	11.6	17.6	19.4
U3 - Toxicity prior Beta(0.3, 0.7)	22.1	17	16	10.9	14.4	19.6
U2 - Toxicity prior Beta(0.5, 0.5)	14.3	13.9	20.9	13.1	19.1	18.7
U3 - Toxicity prior Beta(0.5, 0.5)	21.2	14.4	19.5	10.2	16	18.7
U2 - Toxicity prior Beta(1, 1)	14.8	14.1	20.1	13	19.2	18.8
U3 - Toxicity prior Beta(1, 1)	21.2	13.6	17.8	11.4	17.2	18.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.9	5.1	5.5	3.7	3.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	5	5.4	3.7	3.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	5	5.4	3.7	3.8	
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	5	5.4	3.8	3.7	
U2 - Toxicity prior Beta(0.5, 0.5)	9.1	4.8	5.6	4.1	4.1	
U3 - Toxicity prior Beta(0.5, 0.5)	9.1	4.8	5.7	4.1	4.1	
U2 - Toxicity prior Beta(1, 1)	9.2	4.7	5.6	4	4.2	
U3 - Toxicity prior Beta(1, 1)	9.1	4.7	5.6	4.1	4.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.3	1.2	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.3	1.3	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.8	1.3	1.3	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.8	1.3	1.3	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.6	1.5	1.3	0.6	
U2 - Toxicity prior Beta(1, 1)	0.5	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(1, 1)	0.5	2.6	1.4	1.3	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13	33.4	8.4	2.4	0.8	42
U3 - Toxicity prior Beta(0.15, 0.85)	13.6	34.9	7.3	2	0.8	41.4
U2 - Toxicity prior Beta(0.3, 0.7)	12.6	32.9	9.2	3.4	0.8	41.1
U3 - Toxicity prior Beta(0.3, 0.7)	12.7	33.9	8.3	2.9	0.8	41.4
U2 - Toxicity prior Beta(0.5, 0.5)	12.3	31.6	11.7	3.4	0.9	40.1
U3 - Toxicity prior Beta(0.5, 0.5)	12.4	33.1	10.4	2.7	1.1	40.3
U2 - Toxicity prior Beta(1, 1)	12.3	31.8	11.4	3.3	1	40.2
U3 - Toxicity prior Beta(1, 1)	13.1	33.2	10	2.7	0.9	40.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	13.2	6	3.3	1.5	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	13.2	6	3.3	1.5	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	13	6	3.5	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	13.1	5.9	3.4	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	12.1	6.6	3.8	1.7	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	12.1	6.7	3.8	1.7	0.7	
U2 - Toxicity prior Beta(1, 1)	12.1	6.6	3.8	1.7	0.6	
U3 - Toxicity prior Beta(1, 1)	12	6.7	3.8	1.7	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.3	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.3	0.7	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.2	0.8	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.1	0.9	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.2	0.8	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.2	0.8	0.4	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	66.4	24	2.1	0.1	0	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	68.8	23.1	0.8	0.1	0	7.2
U2 - Toxicity prior Beta(0.3, 0.7)	65.4	25.5	2	0.1	0	7
U3 - Toxicity prior Beta(0.3, 0.7)	68.2	23.8	0.9	0.2	0	6.9
U2 - Toxicity prior Beta(0.5, 0.5)	66.5	24.4	2.2	0.3	0	6.6
U3 - Toxicity prior Beta(0.5, 0.5)	68.9	23	1.2	0.3	0	6.6
U2 - Toxicity prior Beta(1, 1)	66.7	24.1	2.2	0.3	0	6.7
U3 - Toxicity prior Beta(1, 1)	69	22.6	1.2	0.3	0	6.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20	5.8	2.5	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20	5.9	2.4	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	19.9	5.8	2.5	0.6	0	
U3 - Toxicity prior Beta(0.3, 0.7)	19.9	5.9	2.5	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	18.8	6.5	2.8	0.7	0	
U3 - Toxicity prior Beta(0.5, 0.5)	18.9	6.5	2.8	0.8	0	
U2 - Toxicity prior Beta(1, 1)	18.9	6.4	2.8	0.8	0	
U3 - Toxicity prior Beta(1, 1)	18.8	6.5	2.8	0.8	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.3	0.6	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	2.4	0.6	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.3	0.6	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.3	0.6	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.8	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.7	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2.2	0.7	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.7	2.2	0.7	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.2	15.5	12.4	36.6	16.1	3.2
U3 - Toxicity prior Beta(0.15, 0.85)	23	16.3	8.9	34	14.6	3.2
U2 - Toxicity prior Beta(0.3, 0.7)	14.4	15.9	10.6	38.7	17.2	3.2
U3 - Toxicity prior Beta(0.3, 0.7)	21.3	16.6	8.1	34.7	16.1	3.2
U2 - Toxicity prior Beta(0.5, 0.5)	11.8	13.4	10.1	41.3	20.1	3.3
U3 - Toxicity prior Beta(0.5, 0.5)	17.2	14.1	8.3	38.7	18.4	3.3
U2 - Toxicity prior Beta(1, 1)	11.8	13.1	9.9	41.7	20.2	3.3
U3 - Toxicity prior Beta(1, 1)	18	14.2	7.1	39.1	18.3	3.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.3	5.4	5.9	7.4	5.6	
U3 - Toxicity prior Beta(0.15, 0.85)	5.3	5.3	5.9	7.5	5.5	
U2 - Toxicity prior Beta(0.3, 0.7)	5.3	5	5.5	7.8	5.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.2	5.2	5.5	7.7	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	4.7	5.2	8.3	6.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	4.7	5.1	8.4	6.4	
U2 - Toxicity prior Beta(1, 1)	4.9	4.8	5.1	8.4	6.4	
U3 - Toxicity prior Beta(1, 1)	4.9	4.7	5.1	8.4	6.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	1.4	2.3	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	1.4	2.3	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3	1.4	2.2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3	1.4	2.2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.4	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.3	2.9	1.4	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.3	2.9	1.4	2.2	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.2	35	10.3	2.6	0.6	37.3
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	35.7	8.7	2.5	0.4	37.2
U2 - Toxicity prior Beta(0.3, 0.7)	14.7	34.4	11	2.9	0.7	36.3
U3 - Toxicity prior Beta(0.3, 0.7)	15.4	35.2	9.3	3.1	0.5	36.5
U2 - Toxicity prior Beta(0.5, 0.5)	14.1	33.4	13.1	3.5	0.8	35.1
U3 - Toxicity prior Beta(0.5, 0.5)	15.6	33.9	11.8	2.9	0.6	35.2
U2 - Toxicity prior Beta(1, 1)	14.4	32.5	13.2	3.8	0.9	35.2
U3 - Toxicity prior Beta(1, 1)	15.2	34.6	11.3	3.2	0.6	35.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.8	6.5	3.6	1.6	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	12.9	6.4	3.7	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	12.8	6.3	3.8	1.6	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	12.8	6.2	3.7	1.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	11.7	7	4.2	2	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	11.7	6.9	4.2	2	0.7	
U2 - Toxicity prior Beta(1, 1)	11.7	7	4.2	2	0.6	
U3 - Toxicity prior Beta(1, 1)	11.7	7	4.1	2	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.5	0.8	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	1	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.3	1	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.4	1	0.5	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.9	19.9	15.5	24.9	10.2	7.6
U3 - Toxicity prior Beta(0.15, 0.85)	31.4	20.2	10.7	22.4	7.6	7.7
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	20.2	14.6	25.6	11.3	7.6
U3 - Toxicity prior Beta(0.3, 0.7)	31.7	18.1	9.9	24.4	8.5	7.4
U2 - Toxicity prior Beta(0.5, 0.5)	18.1	18.1	14.8	29.1	12.7	7.2
U3 - Toxicity prior Beta(0.5, 0.5)	28.2	19	11	25.9	8.7	7.2
U2 - Toxicity prior Beta(1, 1)	18.9	17.5	15	28.4	12.9	7.3
U3 - Toxicity prior Beta(1, 1)	29.7	18.8	10.3	24.7	9.2	7.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.6	6.1	6	5.5	3.8	
U3 - Toxicity prior Beta(0.15, 0.85)	7.6	6	6.1	5.4	3.8	
U2 - Toxicity prior Beta(0.3, 0.7)	7.3	6	5.8	5.7	4.1	
U3 - Toxicity prior Beta(0.3, 0.7)	7.4	5.8	6	5.8	4.1	
U2 - Toxicity prior Beta(0.5, 0.5)	6.8	5.5	6	6.2	4.4	
U3 - Toxicity prior Beta(0.5, 0.5)	6.8	5.5	6	6.3	4.4	
U2 - Toxicity prior Beta(1, 1)	6.9	5.4	6.1	6.2	4.5	
U3 - Toxicity prior Beta(1, 1)	6.8	5.6	5.9	6.2	4.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.4	1.5	1.9	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.4	1.5	1.8	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.3	1.5	1.9	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.2	1.5	1.9	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.1	1.7	1.9	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3	1.6	1.9	0.7	
U2 - Toxicity prior Beta(1, 1)	0.6	3	1.6	1.9	0.7	
U3 - Toxicity prior Beta(1, 1)	0.5	3.1	1.6	1.9	0.7	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	49.2	30.7	9.3	3.1	0.3	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	58.3	28	4.9	1.2	0.2	7.4
U2 - Toxicity prior Beta(0.3, 0.7)	49.1	30.9	9.4	3.3	0.2	7.1
U3 - Toxicity prior Beta(0.3, 0.7)	59.4	27.3	4.9	1.1	0.2	7.1
U2 - Toxicity prior Beta(0.5, 0.5)	50.5	28	10.3	3.5	0.6	7.1
U3 - Toxicity prior Beta(0.5, 0.5)	60.1	25.3	6.1	1.3	0.2	7
U2 - Toxicity prior Beta(1, 1)	49.9	27.8	10.9	3.9	0.5	7
U3 - Toxicity prior Beta(1, 1)	60.1	25.1	6.2	1.3	0.2	7.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.4	8	5.3	2.5	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	12.4	8	5.3	2.5	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	12	7.7	5.6	2.8	1	
U3 - Toxicity prior Beta(0.3, 0.7)	12.3	7.7	5.4	2.8	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	11.6	7.6	5.5	3.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.6	7.5	5.6	3.3	1.2	
U2 - Toxicity prior Beta(1, 1)	11.6	7.6	5.5	3.3	1.2	
U3 - Toxicity prior Beta(1, 1)	11.6	7.6	5.6	3.3	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.7	1.4	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.6	1.4	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.6	1.5	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.6	1.4	1.1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.3	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.3	1.7	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	1.1	3.3	1.7	1.2	0.3	
U3 - Toxicity prior Beta(1, 1)	1.1	3.3	1.7	1.2	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.3	29.9	15.9	15.8	9	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	35.2	33.4	14.5	11.6	5.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	28.7	28.5	16.7	16.1	9.9	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	34.3	33.9	13.7	13.1	4.9	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	29.5	28.8	15.2	16.1	10.3	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	34.9	33.2	14.8	11.3	5.7	0.1
U2 - Toxicity prior Beta(1, 1)	27.5	30	16	14.7	11.7	0.1
U3 - Toxicity prior Beta(1, 1)	34.1	33.4	14.2	12.3	5.9	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.8	5.9	5.6	6	6.7	
U3 - Toxicity prior Beta(0.15, 0.85)	5.8	5.9	5.5	6	6.7	
U2 - Toxicity prior Beta(0.3, 0.7)	5.8	5.9	5.5	5.9	6.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.8	5.8	5.5	5.9	6.9	
U2 - Toxicity prior Beta(0.5, 0.5)	5.8	5.8	5.5	5.9	7	
U3 - Toxicity prior Beta(0.5, 0.5)	5.8	5.8	5.5	5.9	7.1	
U2 - Toxicity prior Beta(1, 1)	5.8	5.8	5.5	5.9	7.1	
U3 - Toxicity prior Beta(1, 1)	5.7	5.8	5.5	5.8	7.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.2	1.8	2.7	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	1	3.3	1.8	2.7	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.3	1.8	2.7	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	1	3.3	1.8	2.7	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.8	2.6	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.8	2.7	1.4	
U2 - Toxicity prior Beta(1, 1)	1	3.2	1.8	2.7	1.4	
U3 - Toxicity prior Beta(1, 1)	1	3.3	1.8	2.7	1.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.9	54	14.7	1.1	0	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	33.2	54.1	11.9	0.5	0	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	29.7	53.5	15.1	1.4	0	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	32.2	53.9	12.8	0.8	0	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	28.1	53.4	17.2	1	0	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	31.4	53.5	14.3	0.5	0	0.3
U2 - Toxicity prior Beta(1, 1)	28.2	54.1	16.8	0.6	0	0.3
U3 - Toxicity prior Beta(1, 1)	32.3	53.5	13.3	0.6	0	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.8	10.7	6.6	1.6	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	10.9	10.7	6.5	1.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	10.4	6.8	1.8	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	10.5	6.7	1.7	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	10	10.6	7	2.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	10	10.5	7	2.2	0.3	
U2 - Toxicity prior Beta(1, 1)	10	10.6	7	2.2	0.3	
U3 - Toxicity prior Beta(1, 1)	9.9	10.6	7	2.2	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3	1	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.1	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.1	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.2	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	2.8	1.2	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	1.2	0.5	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.4	11.1	11	19.2	48.6	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	14.1	13.4	12.3	17.3	42.2	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	9.2	10.7	10.9	17.9	50.6	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	14.7	12.3	11.8	17.4	43.1	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	9.4	11.2	15.8	54.2	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	13.8	12.1	10.7	16.3	46.4	0.7
U2 - Toxicity prior Beta(1, 1)	8.9	9	11.2	16.8	53.4	0.7
U3 - Toxicity prior Beta(1, 1)	13.7	11.9	11.1	15.2	47.4	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	4.1	5	6	11.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.9	4	5	6	11.1	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	4.1	4.8	5.8	11.4	
U3 - Toxicity prior Beta(0.3, 0.7)	3.8	4.1	4.8	5.9	11.4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	4	4.6	6	11.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	3.9	4.6	6	11.7	
U2 - Toxicity prior Beta(1, 1)	3.7	4	4.6	6	11.6	
U3 - Toxicity prior Beta(1, 1)	3.8	4	4.6	6	11.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.5	2.4	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.4	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.4	2.3	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	1.4	2.4	1.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.2	54.4	13.7	0.6	0	9.1
U3 - Toxicity prior Beta(0.15, 0.85)	24.8	53.2	12.3	0.6	0	9.1
U2 - Toxicity prior Beta(0.3, 0.7)	21.5	54.8	14	0.9	0	8.8
U3 - Toxicity prior Beta(0.3, 0.7)	23.3	54.7	12.5	0.7	0	8.8
U2 - Toxicity prior Beta(0.5, 0.5)	19.4	54.4	16.6	1	0	8.6
U3 - Toxicity prior Beta(0.5, 0.5)	21.3	54.2	15	0.9	0	8.6
U2 - Toxicity prior Beta(1, 1)	19.6	54.5	16	1.3	0	8.6
U3 - Toxicity prior Beta(1, 1)	21.6	54.5	14	1.3	0	8.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	10.9	6	1.4	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	10.9	6.1	1.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.2	10.6	6.2	1.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	10.1	10.7	6.2	1.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	10.9	6.6	2.1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	11	6.6	2	0.2	
U2 - Toxicity prior Beta(1, 1)	9.3	10.9	6.5	2.1	0.2	
U3 - Toxicity prior Beta(1, 1)	9.3	10.8	6.5	2.1	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.2	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.4	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.3	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	1.3	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3.3	1.3	0.6	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.6	49.4	25.7	6.5	0.5	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	16.5	51.5	21.1	6.2	0.4	4.3
U2 - Toxicity prior Beta(0.3, 0.7)	13.4	46.4	27	8.1	0.8	4.3
U3 - Toxicity prior Beta(0.3, 0.7)	17.1	49.3	21.9	6.8	0.6	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	14.3	44	28.3	8.5	0.7	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	18.1	46.1	23.4	7.6	0.6	4.2
U2 - Toxicity prior Beta(1, 1)	13	44.6	28.1	9.5	0.5	4.3
U3 - Toxicity prior Beta(1, 1)	18.5	46.4	22.9	7.2	0.7	4.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.2	9.7	8.1	3.8	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	6.1	9.8	8.2	3.8	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	6.1	9.4	8.1	4.2	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	6	9.4	8.2	4.1	1.6	
U2 - Toxicity prior Beta(0.5, 0.5)	5.7	8.9	8.2	5	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	5.6	8.8	8.2	5	1.7	
U2 - Toxicity prior Beta(1, 1)	5.7	8.8	8.1	5.1	1.7	
U3 - Toxicity prior Beta(1, 1)	5.7	8.9	8.1	5	1.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.8	1.7	1.4	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.8	1.7	1.4	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.7	1.7	1.4	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.7	1.7	1.4	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.9	1.5	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.9	1.5	0.4	
U2 - Toxicity prior Beta(1, 1)	0.4	3.5	1.9	1.5	0.4	
U3 - Toxicity prior Beta(1, 1)	0.4	3.5	1.9	1.5	0.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	58	1.6	0	0	0	40.4
U3 - Toxicity prior Beta(0.15, 0.85)	58.1	1.6	0	0	0	40.3
U2 - Toxicity prior Beta(0.3, 0.7)	58.1	1.6	0	0	0	40.3
U3 - Toxicity prior Beta(0.3, 0.7)	58.5	1.6	0	0	0	39.9
U2 - Toxicity prior Beta(0.5, 0.5)	59.1	1.3	0	0	0	39.6
U3 - Toxicity prior Beta(0.5, 0.5)	59	1.2	0	0	0	39.8
U2 - Toxicity prior Beta(1, 1)	58.4	1.5	0	0	0	40.1
U3 - Toxicity prior Beta(1, 1)	58.9	1.4	0	0	0	39.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	21.4	1.7	0.3	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	21.4	1.6	0.3	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	21.4	1.7	0.3	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	21.4	1.7	0.3	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	21.3	1.8	0.3	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	21.3	1.8	0.3	0	0	
U2 - Toxicity prior Beta(1, 1)	21.2	1.8	0.3	0	0	
U3 - Toxicity prior Beta(1, 1)	21.3	1.8	0.3	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.6	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.5	58.8	10.6	0.5	0	4.6
U3 - Toxicity prior Beta(0.15, 0.85)	28.5	57.8	8.8	0.4	0	4.5
U2 - Toxicity prior Beta(0.3, 0.7)	24.4	59.5	10.5	1.1	0	4.5
U3 - Toxicity prior Beta(0.3, 0.7)	29.3	55.7	9.2	1.3	0	4.5
U2 - Toxicity prior Beta(0.5, 0.5)	23.1	57.2	13.8	1.5	0	4.4
U3 - Toxicity prior Beta(0.5, 0.5)	27.3	56	10.9	1.3	0	4.5
U2 - Toxicity prior Beta(1, 1)	23.5	58	12.8	1.1	0	4.6
U3 - Toxicity prior Beta(1, 1)	27.2	56.3	11	1.1	0	4.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.9	12.9	5.3	1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	12.9	5.3	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	12.8	5.4	1.2	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.8	12.5	5.6	1.2	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	9	12.8	5.6	1.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	12.9	5.6	1.8	0.2	
U2 - Toxicity prior Beta(1, 1)	9	12.7	5.5	1.9	0.2	
U3 - Toxicity prior Beta(1, 1)	9	12.7	5.6	1.9	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.1	1.1	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.1	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4	1.1	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.2	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.4	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.3	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	1.3	0.4	0	
U3 - Toxicity prior Beta(1, 1)	0.7	3.7	1.3	0.4	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17	6	0	0	0	77
U3 - Toxicity prior Beta(0.15, 0.85)	17.2	5.5	0	0	0	77.3
U2 - Toxicity prior Beta(0.3, 0.7)	17	5.5	0	0	0	77.5
U3 - Toxicity prior Beta(0.3, 0.7)	17.8	5.7	0	0	0	76.5
U2 - Toxicity prior Beta(0.5, 0.5)	18.4	4.2	0	0	0	77.4
U3 - Toxicity prior Beta(0.5, 0.5)	19.1	3.6	0	0	0	77.3
U2 - Toxicity prior Beta(1, 1)	18.4	4.3	0	0	0	77.3
U3 - Toxicity prior Beta(1, 1)	19.3	3.6	0	0	0	77.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.4	2.1	0.3	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	17.4	2.1	0.3	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.5	2.2	0.3	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	17.5	2.1	0.3	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.4	2.5	0.3	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	17.4	2.5	0.3	0	0	
U2 - Toxicity prior Beta(1, 1)	17.4	2.4	0.3	0	0	
U3 - Toxicity prior Beta(1, 1)	17.4	2.5	0.3	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	1.4	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	1.3	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	1.4	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	1.3	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.3	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.3	1.3	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.3	1.3	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	1.5	5.6	52.4	39.3	0
U3 - Toxicity prior Beta(0.15, 0.85)	2	2.9	6.7	53.5	34.9	0
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	1.5	5.3	52.9	39.2	0
U3 - Toxicity prior Beta(0.3, 0.7)	2	2.8	6.1	54.4	34.7	0
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	1.7	5.1	51.3	40.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	2.2	3	6.1	53.1	35.6	0
U2 - Toxicity prior Beta(1, 1)	1.2	1.6	5.3	51.3	40.6	0
U3 - Toxicity prior Beta(1, 1)	1.7	2.8	6.7	54.2	34.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.2	3.2	3.8	9	10.8	
U3 - Toxicity prior Beta(0.15, 0.85)	3.2	3.1	3.8	9	10.8	
U2 - Toxicity prior Beta(0.3, 0.7)	3.2	3.2	3.8	8.9	10.9	
U3 - Toxicity prior Beta(0.3, 0.7)	3.2	3.1	3.8	9	10.9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.2	3.1	3.7	8.9	11	
U3 - Toxicity prior Beta(0.5, 0.5)	3.2	3.2	3.7	8.9	11	
U2 - Toxicity prior Beta(1, 1)	3.2	3.1	3.7	9	11	
U3 - Toxicity prior Beta(1, 1)	3.2	3.1	3.7	9	10.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.3	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.3	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.3	1.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	32.8	35.8	26.6	4.2	0	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	39.3	33.6	23	3.4	0	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	32.6	34.2	28.6	3.9	0	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	38.5	33.8	23.7	3.3	0	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	32.6	31.6	31.1	4.1	0	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	37.5	32.7	25.7	3.5	0	0.6
U2 - Toxicity prior Beta(1, 1)	32.3	31.6	31.8	3.7	0	0.6
U3 - Toxicity prior Beta(1, 1)	38	31.2	26.6	3.6	0	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	9.3	8.2	2	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	10.1	9.2	8.3	2	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	8.9	8.7	2.3	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9.8	8.8	8.7	2.3	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	9.3	8.5	8.9	2.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	8.5	8.9	2.9	0.3	
U2 - Toxicity prior Beta(1, 1)	9.2	8.5	8.9	2.9	0.3	
U3 - Toxicity prior Beta(1, 1)	9.3	8.4	8.9	2.9	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.3	1.4	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.3	1.5	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.2	1.5	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.3	1.5	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.1	1.6	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.1	1.6	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.1	1.6	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.1	1.7	0.7	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31.1	32.4	21.8	3.8	0	10.9
U3 - Toxicity prior Beta(0.15, 0.85)	34.2	33.3	18.7	2.9	0	10.9
U2 - Toxicity prior Beta(0.3, 0.7)	30.9	31.4	22.5	4.4	0	10.8
U3 - Toxicity prior Beta(0.3, 0.7)	33.1	33	20	3.4	0	10.5
U2 - Toxicity prior Beta(0.5, 0.5)	28.3	31	25.9	4.6	0	10.2
U3 - Toxicity prior Beta(0.5, 0.5)	31.8	30.4	23.3	4.2	0	10.3
U2 - Toxicity prior Beta(1, 1)	27.7	30.4	27.1	4.6	0	10.2
U3 - Toxicity prior Beta(1, 1)	31.8	30.2	23.1	4.7	0	10.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.2	7.7	6.7	1.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	12.2	7.8	6.6	1.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	12	7.3	7	2.2	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	12	7.2	7.1	2.1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	11.2	7.3	7.3	2.7	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	11.3	7.2	7.5	2.6	0.3	
U2 - Toxicity prior Beta(1, 1)	11.2	7.2	7.5	2.6	0.3	
U3 - Toxicity prior Beta(1, 1)	11.3	7.2	7.5	2.6	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.1	0.7	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.1	0.7	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.1	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.1	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.3	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.3	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.8	1.3	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.8	1.3	0.7	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.8	28.8	37.3	6.4	0.2	5.5
U3 - Toxicity prior Beta(0.15, 0.85)	25.3	30.1	33.9	5.1	0.1	5.5
U2 - Toxicity prior Beta(0.3, 0.7)	22.4	26.5	37.7	7.8	0.2	5.4
U3 - Toxicity prior Beta(0.3, 0.7)	25	28	34.6	6.9	0.1	5.4
U2 - Toxicity prior Beta(0.5, 0.5)	20.4	25	40.1	8.9	0.2	5.4
U3 - Toxicity prior Beta(0.5, 0.5)	24.6	25.9	35.9	8.1	0.1	5.4
U2 - Toxicity prior Beta(1, 1)	20.4	24.7	40.8	8.5	0.2	5.4
U3 - Toxicity prior Beta(1, 1)	23.6	26.3	36.8	7.8	0.1	5.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	7.4	9	3.1	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	8.8	7.5	9	3.2	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	8.7	7	9.2	3.5	1	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	7	9.1	3.5	1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.2	6.5	9.4	4.3	1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	6.5	9.3	4.3	1	
U2 - Toxicity prior Beta(1, 1)	8.2	6.4	9.5	4.3	1	
U3 - Toxicity prior Beta(1, 1)	8.2	6.5	9.3	4.4	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.6	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.5	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.5	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.1	1.5	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.9	1.7	1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.9	1.7	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	1.7	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.4	2.9	1.7	1	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.8	38.7	3	0	0	36.5
U3 - Toxicity prior Beta(0.15, 0.85)	22.4	38.2	2.9	0	0	36.5
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	39.5	4.1	0	0	35.7
U3 - Toxicity prior Beta(0.3, 0.7)	22.4	39.1	2.9	0	0	35.6
U2 - Toxicity prior Beta(0.5, 0.5)	21.6	39	4.6	0.1	0	34.7
U3 - Toxicity prior Beta(0.5, 0.5)	22.2	39.3	3.7	0.2	0	34.6
U2 - Toxicity prior Beta(1, 1)	20.7	40	4.4	0.2	0	34.7
U3 - Toxicity prior Beta(1, 1)	22.7	38.8	3.4	0.1	0	35
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	13.6	8.1	2.9	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	13.6	8.1	3	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	13.5	8.1	3.1	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	13.5	8.1	3.1	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	12.5	9	3.4	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	12.4	9.2	3.3	0.7	0	
U2 - Toxicity prior Beta(1, 1)	12.5	9	3.4	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	12.5	9	3.4	0.7	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	0.6	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	0.6	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	0.7	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	0.6	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	0.8	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	0.8	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.4	2.5	0.8	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.3	2.5	0.8	0.2	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	67.9	26.7	2.4	0	0	3
U3 - Toxicity prior Beta(0.15, 0.85)	69.8	24.9	2.2	0	0	3.1
U2 - Toxicity prior Beta(0.3, 0.7)	67.2	27.2	2.6	0	0	3
U3 - Toxicity prior Beta(0.3, 0.7)	69	25.9	2.1	0	0	3
U2 - Toxicity prior Beta(0.5, 0.5)	67.2	27.3	2.9	0	0	2.6
U3 - Toxicity prior Beta(0.5, 0.5)	70.6	24.9	2	0.1	0	2.4
U2 - Toxicity prior Beta(1, 1)	67.7	27	2.6	0	0	2.7
U3 - Toxicity prior Beta(1, 1)	70.7	24.4	2.3	0	0	2.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.2	7.3	2.7	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	19	7.4	2.7	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	18.9	7.4	2.8	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	19	7.3	2.8	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	18.1	8	2.8	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	18.2	7.9	2.9	0.5	0	
U2 - Toxicity prior Beta(1, 1)	18.2	7.9	2.9	0.5	0	
U3 - Toxicity prior Beta(1, 1)	18.2	7.9	2.9	0.5	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.1	0.7	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.1	0.6	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.7	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.7	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2	0.7	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.7	2	0.7	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19	19.6	16.6	35.9	8.1	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	25.1	22.9	13.3	31.8	6.1	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	17.9	18.1	16.1	37.9	9.2	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	25.2	21.4	13.8	31.7	7.1	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	16.7	16.7	14.8	40.1	10.9	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	22.3	20.6	13.8	33.5	9	0.8
U2 - Toxicity prior Beta(1, 1)	16.5	17.1	14.8	39.8	11	0.8
U3 - Toxicity prior Beta(1, 1)	23.3	19.2	12.9	34.3	9.5	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.4	5.1	6.4	8.1	4.9	
U3 - Toxicity prior Beta(0.15, 0.85)	5.4	5.1	6.4	8.1	4.9	
U2 - Toxicity prior Beta(0.3, 0.7)	5.3	4.9	6.1	8.4	5.2	
U3 - Toxicity prior Beta(0.3, 0.7)	5.3	4.9	6.1	8.4	5.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.1	4.7	5.8	8.9	5.4	
U3 - Toxicity prior Beta(0.5, 0.5)	5.1	4.6	5.8	9	5.4	
U2 - Toxicity prior Beta(1, 1)	5.1	4.6	5.8	9	5.4	
U3 - Toxicity prior Beta(1, 1)	5.1	4.6	5.8	8.9	5.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.7	2.3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.7	2.3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.6	2.3	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.6	2.3	0.9	
U2 - Toxicity prior Beta(1, 1)	0.4	2.8	1.6	2.3	0.9	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	1.6	2.3	0.9	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.6	35.9	27.6	24.7	6.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	8.4	42.2	25.2	19.9	4.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	5.8	35.2	28.3	24.8	5.8	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	8	42	26.1	19.9	3.9	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	5.3	34.7	29.2	24.2	6.5	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	7.6	42	26.1	19.6	4.6	0.1
U2 - Toxicity prior Beta(1, 1)	5.8	35.7	28.2	23.6	6.6	0.1
U3 - Toxicity prior Beta(1, 1)	7.9	42.6	25.7	18.8	4.9	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	5.9	6.7	8	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	5.9	6.6	8.1	5.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	5.8	6.5	8.1	5.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	5.9	6.5	8	5.5	
U2 - Toxicity prior Beta(0.5, 0.5)	4	5.9	6.4	8	5.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4	5.9	6.4	8	5.7	
U2 - Toxicity prior Beta(1, 1)	4	6	6.3	8	5.7	
U3 - Toxicity prior Beta(1, 1)	4	5.9	6.4	8	5.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.8	1.4	1.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	16.5	33.5	30.3	9.5	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	14.5	21.4	32.7	25.4	5.4	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	16.8	32.9	29.5	11.1	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	15.8	19.9	31.9	26.1	5.7	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	16.3	32.1	30.3	12	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	14.6	20	33.4	25.1	6.3	0.6
U2 - Toxicity prior Beta(1, 1)	9	15.4	32.6	30.5	11.9	0.6
U3 - Toxicity prior Beta(1, 1)	14.5	19.4	32.8	25.5	7.2	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.9	4.5	7.9	7.7	6	
U3 - Toxicity prior Beta(0.15, 0.85)	4	4.5	7.8	7.7	6	
U2 - Toxicity prior Beta(0.3, 0.7)	3.9	4.4	7.6	7.8	6.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4	4.4	7.5	7.9	6.2	
U2 - Toxicity prior Beta(0.5, 0.5)	3.9	4.4	7.4	7.9	6.4	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	4.4	7.4	7.9	6.4	
U2 - Toxicity prior Beta(1, 1)	4	4.3	7.3	7.9	6.4	
U3 - Toxicity prior Beta(1, 1)	3.9	4.4	7.5	7.8	6.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2	2.7	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2	2.7	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2	2.7	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2	2.7	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	2	2.6	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	2	2.6	1.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.9	2	2.7	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.9	2	2.7	1.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7	8	15	44.5	25.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	13	9.7	14.3	42.3	20.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	7.4	13.2	43.6	28.3	0
U3 - Toxicity prior Beta(0.3, 0.7)	12.9	8.8	14.4	41.9	22	0
U2 - Toxicity prior Beta(0.5, 0.5)	7	5.7	11.2	45.4	30.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	8.5	12.8	44	24.2	0
U2 - Toxicity prior Beta(1, 1)	6.5	6.4	10.4	45.8	30.9	0
U3 - Toxicity prior Beta(1, 1)	9.9	8.6	12.5	44.6	24.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.7	5.5	8.6	8.7	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.7	5.5	8.6	8.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	3.6	5.2	8.6	9.1	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.7	5.2	8.7	9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	3.6	5	8.9	9.1	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.6	5	8.9	9.1	
U2 - Toxicity prior Beta(1, 1)	3.4	3.6	5	8.9	9.2	
U3 - Toxicity prior Beta(1, 1)	3.4	3.6	5	8.9	9.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.4	2.8	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.4	2.8	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.8	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.8	16.6	49	13.2	3.4	2
U3 - Toxicity prior Beta(0.15, 0.85)	21.3	18.4	43.8	12.2	2.3	2
U2 - Toxicity prior Beta(0.3, 0.7)	16.4	16.3	45.5	16.2	3.6	2
U3 - Toxicity prior Beta(0.3, 0.7)	22.2	18.3	40.7	14.3	2.5	2
U2 - Toxicity prior Beta(0.5, 0.5)	14.1	16.2	42.6	21.5	3.7	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	19.7	18	40.1	17.4	2.9	1.9
U2 - Toxicity prior Beta(1, 1)	13.4	14.7	44.9	20.5	4.6	1.9
U3 - Toxicity prior Beta(1, 1)	20.1	18.1	38.9	18	3	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.7	6.7	10.1	5	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	5.7	6.6	10.2	5	2.2	
U2 - Toxicity prior Beta(0.3, 0.7)	5.6	6.3	10	5.3	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	5.5	6.3	10	5.3	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	5.2	5.9	9.4	6.4	2.8	
U3 - Toxicity prior Beta(0.5, 0.5)	5.1	5.9	9.5	6.4	2.8	
U2 - Toxicity prior Beta(1, 1)	5.2	5.9	9.4	6.4	2.8	
U3 - Toxicity prior Beta(1, 1)	5.2	5.9	9.5	6.4	2.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.4	2	1.5	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.4	2	1.5	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.9	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	2	1.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.1	2	1.6	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.1	2.1	1.6	0.6	
U2 - Toxicity prior Beta(1, 1)	0.5	3.1	2.1	1.6	0.6	
U3 - Toxicity prior Beta(1, 1)	0.5	3.1	2.1	1.6	0.6	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.4	45.1	31.2	10.6	1.6	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	14.8	49.2	26.3	8.3	1.3	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	44.4	32.1	10.8	1.8	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	14.2	48.2	27.1	9.1	1.3	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	10.7	43.3	30.9	12.7	2.3	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	13.5	47.8	25.8	11.1	1.7	0.1
U2 - Toxicity prior Beta(1, 1)	10.5	44.1	30.6	12.5	2.2	0.1
U3 - Toxicity prior Beta(1, 1)	14.2	47.4	26.1	10.6	1.6	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	9.1	9	4.9	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	9.1	9.1	4.9	2.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	8.6	8.9	5.2	2.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	8.8	8.8	5.2	2.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.4	8.3	8.4	6.1	2.8	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	8.3	8.5	6	2.8	
U2 - Toxicity prior Beta(1, 1)	4.4	8.3	8.4	6.1	2.8	
U3 - Toxicity prior Beta(1, 1)	4.5	8.3	8.5	6	2.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.2	1.6	1.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.2	1.6	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.2	1.6	1.3	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.2	1.6	1.3	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.7	1.4	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.7	1.4	0.5	
U2 - Toxicity prior Beta(1, 1)	0.3	3.1	1.7	1.4	0.5	
U3 - Toxicity prior Beta(1, 1)	0.3	3.1	1.7	1.4	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.1	46.8	11.2	2.8	0.9	16.2
U3 - Toxicity prior Beta(0.15, 0.85)	25.8	45.3	9.4	2.8	0.5	16.2
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	46.6	11.9	4.4	0.6	15.8
U3 - Toxicity prior Beta(0.3, 0.7)	25	45.5	9.2	4	0.4	15.9
U2 - Toxicity prior Beta(0.5, 0.5)	20.7	44.2	13.9	5	0.9	15.3
U3 - Toxicity prior Beta(0.5, 0.5)	26	42.7	11.2	4.4	0.5	15.2
U2 - Toxicity prior Beta(1, 1)	21.3	43.5	13.6	5.2	1	15.4
U3 - Toxicity prior Beta(1, 1)	24.8	43.3	11.2	4.7	0.6	15.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.1	8.9	5	2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	11.1	8.8	5	2	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	8.8	5	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	8.9	4.9	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	9.1	5.3	2.7	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	9.9	9.1	5.3	2.7	0.9	
U2 - Toxicity prior Beta(1, 1)	10	9	5.2	2.7	0.9	
U3 - Toxicity prior Beta(1, 1)	9.9	9.1	5.2	2.7	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	1.1	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	1.1	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.3	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.3	1.3	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	3.3	1.3	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.5	3.3	1.3	0.7	0.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.6	56.6	9.3	1.3	0.1	7.1
U3 - Toxicity prior Beta(0.15, 0.85)	32.3	52.6	7	0.8	0.1	7.2
U2 - Toxicity prior Beta(0.3, 0.7)	24.2	55.1	11.5	2.1	0.2	6.9
U3 - Toxicity prior Beta(0.3, 0.7)	30	53.3	8.4	1.4	0.1	6.8
U2 - Toxicity prior Beta(0.5, 0.5)	22.7	54	14	2.5	0.2	6.6
U3 - Toxicity prior Beta(0.5, 0.5)	30.4	49.5	11.3	2.1	0.1	6.6
U2 - Toxicity prior Beta(1, 1)	25.2	52.6	12.8	2.3	0.3	6.8
U3 - Toxicity prior Beta(1, 1)	30.5	50.3	9.9	2.3	0.1	6.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	11	5.6	1.6	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	11	5.7	1.6	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	10.9	5.7	1.9	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.7	10.9	5.7	1.9	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	11.1	5.8	2.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	11	5.8	2.4	0.6	
U2 - Toxicity prior Beta(1, 1)	8.9	11	5.8	2.4	0.6	
U3 - Toxicity prior Beta(1, 1)	8.9	11	5.8	2.4	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4	1.2	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.4	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.5	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	1.5	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.7	3.7	1.5	0.7	0.2	